Assignment 1
徐亿甜 124090767

About Data:
1. Monthly Return: Monthly Closing Price, Market Value of Tradable Shares, Monthly Return with Cash Dividend Reinvestment
2. Index: EPS_TTM1, Net Assets per share, Total Assets, Total Liabilities, ROA_TTM, ROE_TTM, R&D Expenses
3. Company Basic Information: Establish Date, Market Type

I chose EPS_TTM1, because it reflects a firm's earnings over the most recent twelve months, which smooths quarterly fluctuations and provides a more stable and commonly used measure for calculating the P/E ratio. To match the EPS, I chose ROA_TTM, ROE_TTM as well.

1b.

	Market_Group	Count	Mean	Median	Std	P25	P75
Firm_Age	GEM Board	131111	17.484921	17.264887	5.556165	13.519507	21.026694
Firm_Age	Main Board	571423	18.137437	17.932923	7.420658	12.695414	23.123888
Mretwd	GEM Board	129698	0.009519	-0.006162	0.163476	-0.082999	0.078211
Mretwd	Main Board	569970	0.012162	-0.001158	0.156604	-0.067905	0.074667
PB_Ratio	GEM Board	131111	4.391793	3.237887	19.572903	2.200067	5.090415
PB_Ratio	Main Board	571415	2.153044	2.441319	668.2435	1.534867	4.004402
PE_Ratio	GEM Board	131111	84.27988	40.465815	9646.8959	22.811393	70.610916
PE_Ratio	Main Board	571423	38.84603	27.295401	6100.0643	12.473397	55.312776
RD_Ratio	GEM Board	91413	0.02114	0.015118	0.025511	0.007597	0.026855
RD_Ratio	Main Board	193940	0.012769	0.008233	0.015694	0.002755	0.017737
ROA	GEM Board	127249	0.040805	0.044392	0.075816	0.013133	0.077497
ROA	Main Board	555529	0.0332	0.03216	0.296276	0.008129	0.065603
ROE	GEM Board	127030	0.033827	0.064015	1.689861	0.020047	0.109053
ROE	Main Board	550610	0.052582	0.063057	3.688257	0.017416	0.115781

1. Return & Volatility

The mean return of Main Board firms is slightly higher than GEM Board firms. However, the median returns are much lower than the mean in both markets, indicating that the distribution of stock returns is right-skewed. This suggests that a small number of extreme positive returns pull the mean upward. Investors should highly be wary of price fluctuations and market bubbles.

2. **Valuation**

(To ensure the validity of the PE ratio, I first filtered the EPS data by removing observations with non-positive values. Then, I removed extreme EPS observations using the 1st and 99th percentile-based approach. After applying these filters, the PE ratio is calculated, and selected with same approach. This procedure helps getting a clean and economically meaningful data-set for subsequent analysis.)

GEM Board firms generally exhibit higher market valuations than those on the Main Board. These results suggest that investors tend to assign higher valuation multiples to firms listed on the GEM Board. This is likely because GEM firms are

typically younger and more growth-oriented, leading to stronger market expectations regarding their future growth and profitability.

3. **Profitability**

For ROE, the Main Board firms exhibit a higher average value compared with GEM firms, indicating higher returns to shareholders. The median ROA is slightly higher for GEM firms as well, suggesting somewhat stronger asset profitability among GEM firms.This may reflect the fact that Main Board firms are generally larger and more mature, leading to higher but also greater variation in return. And GEM Board firms are in growth stage, showing greater profit margin from assets.

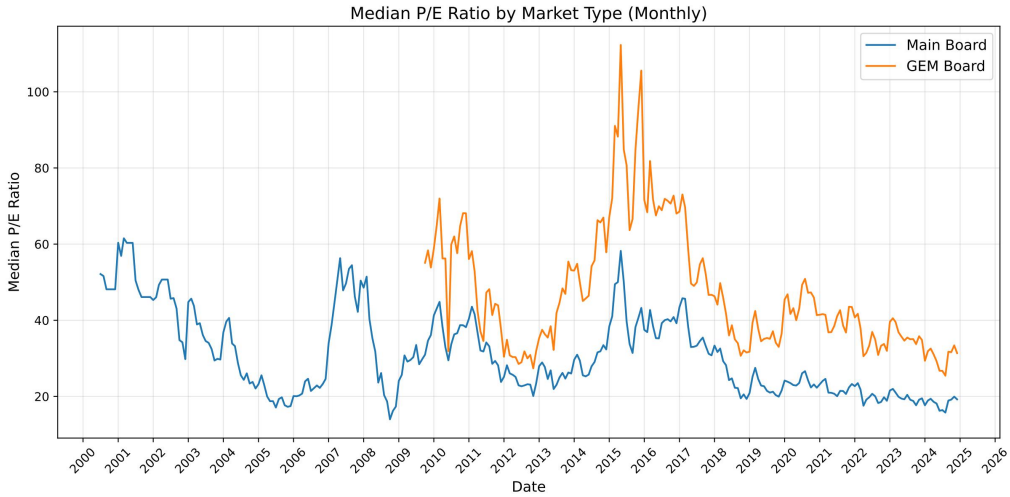
4. Innovation

GEM Board firms exhibit higher R&D intensity compared with those on the Main Board. These results suggest that GEM firms tend to invest more heavily in research and development, which is consistent with the fact that the GEM market primarily hosts younger and more innovation-oriented firms.

5. Firm Age

For firm age, companies listed on the Main Board are slightly older than those on the GEM Board. This suggests that Main Board firms tend to be more mature, while GEM firms are relatively younger.

2.



(1) PE: GEM > Main, Stability: Main > GEM.

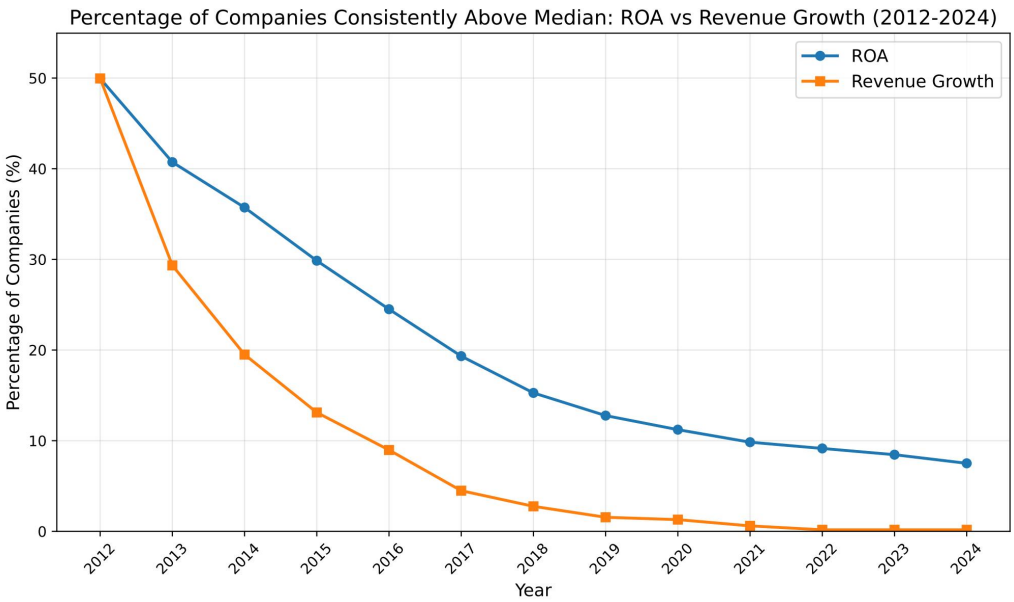
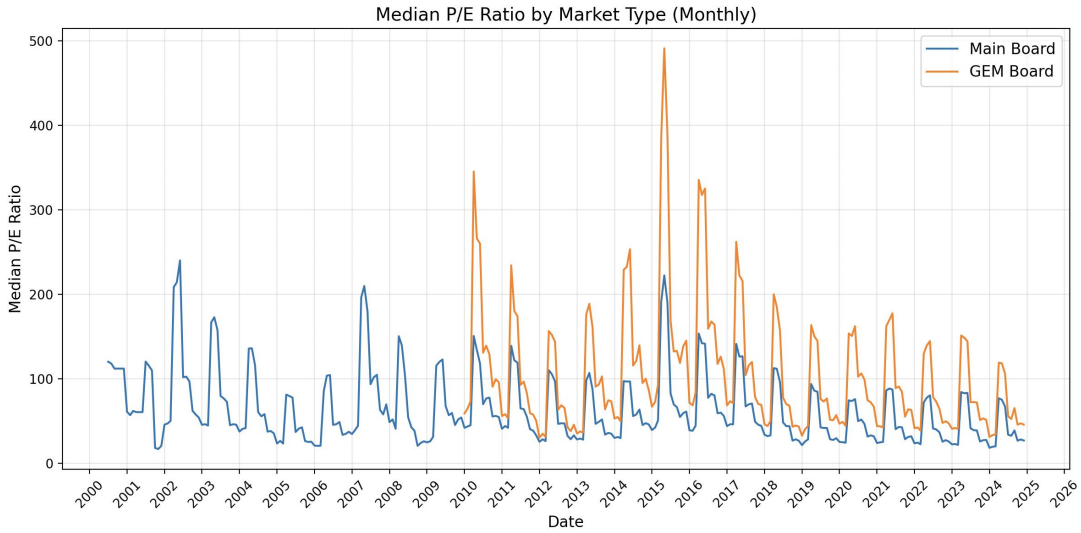
The figure shows that GEM Board consistently exhibits higher and more volatile P/E ratios than the Main Board. However, the GEM Board experiences larger fluctuations in P/E ratios over time. For instance, around 2014–2016 the GEM Board's P/E ratio increased sharply and reached extremely high level and sharply fell down, suggesting greater volatility and stronger sensitivity to market sentiment. These big changes may considered as market bubbles. In contrast, the Main Board maintains relatively lower and more stable P/E ratios, implying a more mature and stable market environment.

In recent years, the P/E ratios of both markets have gradually declined and remain at relatively low levels by Dec.2024. This relatively low valuation may provide a potential opportunity for investors to consider making new investments in either

market. The investors should consider whether it is a signal of undervalued and predict the future direction. For investors who are risk- adverse, Main Board market suits them better with smaller fluctuation range in valuation. For investors seeking higher potential returns, GEM Board may offer opportunities, but also much higher risk.

(2) Index ETF, different from individual stocks, tracks the trend of the whole sector, representing the overall market valuation with lower risk and low trading cost. A possible trading strategy is adjusting market exposure based on valuation signals, thus, the P/E ratio. When the market P/E ratio is relatively low compared with its historical level, the market may be undervalued, suggesting higher expected future returns. In this case, an investor could buy or increase holdings of a broad market index ETF to gain exposure to the potential market rebound. Conversely, when the market P/E ratio rises significantly above its historical average, the market may be overvalued, and the investor could reduce the ETF position or move funds to safer assets. For example, when the market P/E ratio falls below the 25th percentile of its historical distribution, the investor may buy or increase holdings of a broad market index ETF. Conversely, when the market P/E ratio rises above the 75th percentile, the investor may sell the ETF or reduce the position to lock in profits and lower exposure to market risk.

3.



Add: problem 2 picture with not TTM data